

Precision Pouring into Unknown Containers by Service Robots

Chenyu Dong, Masaru Takizawa, Shunsuke Kudoh, and Takashi Suehiro

Abstract—Service robots have developed considerably in recent years. This study aims to propose two approaches for controlling the motion of a service robot as it pours liquid precisely from an unknown container into another unknown container without the need of any external tools. To realize this task, we must resolve two sub-tasks, which are determining the poured volume and controlling the pouring action. Our first proposal concentrates on the target container. The poured volume is calculated using a model of target container and the height of liquid in the target container. The action is controlled using a proportional-derivative controller, which considers the angular speed of the pouring container as a process variable and the poured volume as a control variable. Our second method concentrates on the pouring container. The poured volume is calculated using the relation between the angle of the pouring container and poured volume. The action is controlled with a simple proportional controller that takes the angular speed of the pouring container as a process variable and target angle as a control variable. A point cloud is used to model the two containers. These two methods were implemented in a dual-arm robot system for testing, and the results show that both methods are effective for controlling precise pouring tasks.

I. INTRODUCTION

Household service robots need to perform a wide variety of functions, such as cooking, personal care, sweeping, and organization, among others. This study concentrates on a specific pouring task, which is involved in many cooking and caring tasks. In daily life, people typically use measurement tools, such as a scale, measuring cup, or measuring spoon, to ensure that a precise amount of liquid is poured. However, these measurement tools increase the time and number of steps needed to execute the task. For example, when using measurement tools to separate liquid into several cups, the volume of the liquid in each cup must be measured, which increase the time needed for the task. However, unlike humans, robots could be designed to pour a precise volume with a camera or other sensors, eliminating the need of separate measurement tools. To achieve this functionality, two problems must be addressed, which are measuring the poured volume and controlling the pouring action.

The poured volume determines the requirements for the pouring action; therefore, calculating the poured volume using sensor inputs is a key step in controlling the motion. With a precise measurement of the poured volume, the motion can be controlled. Several studies have reported

methods for controlling the precise pouring task. For liquid detection, convolutional neural networks (CNNs) have been used to detect the type of container and liquid volume in that container from an RGB image [1]. However, the output from the CNN is restricted to ten classes of volumes, which are not fine-grained enough for the precise pouring task. The height of the liquid in the target container has been detected using the refractive index of the liquid [2]. The region of liquid in a color image can also be used [3], [4]. For controlling the pouring action, reinforcement learning has been used to guide a robot arm for pouring water into a container [5]. Imitation learning has also been used [6]. The trajectory for pouring can be encoded using an hidden Markov model, and then can be reproduced using Gaussian mixture regression [7]. The demonstration method has also been used to teach the robot to pour liquid [8]. The trajectory of the pouring container can be modeled using parameters such as the position or size of the target container [9]. Also, using neural networks to generate the pouring trajectory is present in [10].

For controlling the pouring task, an RGB-D image and thermal image has been used to pour hot water [11]. The RGB-D and thermal images are input to a CNN to estimate the poured volume and control the poured volume with a simple PID-controller. This method is accurate within 38 ml. However, in some cases such as cooking, 38 ml is not sufficiently precise. The pouring task has also been realized by specifying a liquid height instead of a target volume [12]. A few studies have focused on modeling the pouring task [13], [14]. The precise pouring task has been implemented with a cube-shaped pouring container and a cylindrical target container [13]. That work was improved upon with a symmetric pouring container [14].

This paper builds on this body of work with the following contributions:

- 1) No external measurement tools are used. The proposed design achieves precise pouring using only sensor inputs from a camera on the robot wrist.
- 2) Only the shape of one of containers (the pouring container or the target container) is required.
- 3) The proposed approach uses a point cloud to model the containers; therefore, it is not sensitive to the shape of the container.

In one method, an external scale and chess board under the target container are required [14]. Some methods also require that the shape of both sides containers is known [13], [14]. One of the methods was not sufficiently precise for pouring tasks in cooking [11]. The [12] can only pour the liquid to

C. Dong, S. Kudoh and T. Suehiro are from the Graduate School of Informatics and Engineering at the University of Electro-Communications, 1-5-1, Chofugaoka, Chofu, Tokyo, Japan (tou@taka.is.uec.ac.jp, s-kudoh@uec.ac.jp, suehiro@is.uec.ac.jp).

M. Takizawa is from the Graduate School of Information Systems at the University of Electro-Communications, 1-5-1, Chofugaoka, Chofu, Tokyo, Japan (takizawa@taka.is.uec.ac.jp).

a specified height, and it also require the shape of cup is cylinder, because of using a cylinder model to match the cup.

In liquid detection, since the proposed method infers the geometry of the container and the liquid level from the point-cloud model, the accuracy of the volume measurement can be improved [1]. For controlling the pouring action such as in liquid detection, the proposed method measures the position and geometry of the container; therefore, the method can focus entirely on the pouring action instead of needing to generate an entire trajectory using reinforcement or imitation learning [5], [6], [7], [8] or parameter based method [9].

The rest of this paper is organized as follows: Section II-C, II-D describe the two main proposed methods and Section II-B discusses the volume calculations used in the two methods. Next, Section III-A discusses the hardware that is used to obtain the point cloud of a complete container. Then, Sections III-B, III-C and III-D present the details of how the two methods were implemented. Finally, tests for each method are described in Section IV.

II. METHODOLOGY

A. Overview

Any pouring task involves pouring and target containers. Therefore, we propose two pouring methods that focus on each of these containers. Method 1 focuses on the target container. For a normal liquid with typical viscosity, such as water or milk, among others, the liquid level always forms a horizontal plane. According to this, we can assume that the volume of container under the liquid-level plane is the poured volume. The action control uses a proportional-derivative (PD) controller that takes the angular speed of the pouring container as a process variable and poured volume as a control variable.

Method 2 concentrates on the pouring container. Under ideal conditions, the volume poured out of the pouring container is always equal to the volume poured into the target container. Therefore, if the relation between the poured volume and rotation angle of the pouring container can be calculated, the pouring task is simply rotating the pouring container. The controller used in Method 2 is simple with only a proportional (P) controller that takes the angular speed of the pouring container as a process variable and rotation angle of target volume as a control variable.

B. Calculation of Volume

For either method, the partial volume of the container model must be calculated, such as the volume of liquid under the liquid level. In our approach, for any convex-shaped container with a constant rotation angle θ^1 , the calculation volume V_H from the bottom of the container to the height H is marked in blue in Figure 1.

The volume of $V_{H,\theta}$ is calculated by

$$V_{H,\theta} = \int_0^H S(h,\theta)dh \quad (1)$$

¹To be precise, this is the pose of the container, but for simplicity in the description, we use the angle instead of the pose.

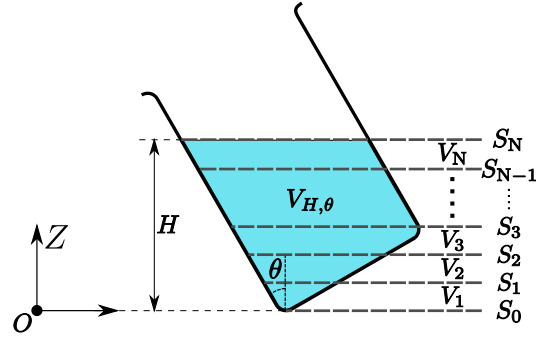


Fig. 1: V_H is the volume that we want to calculate. H is the height, and can be set between the bottom of the model and top of the model. To calculate the integration, $S_0, S_1, S_2, \dots, S_N$ are the areas of each cross-sectional plane. $V_1, V_2, V_3, \dots, V_N$ are the volumes of each slice part.

where $S(h,\theta)$ is the cross-sectional area that is cut by the plane at height h , and the model has a constant rotation angle θ . $S(h,\theta)$ is a function of the container model with the variables θ and h .

The integration mentioned above is calculated as follows. Herein, we use a point cloud to model the container. First, we slice the model into several parts, and the method for slicing is discussed in Section III-C). As shown in Figure 1, the container model is sliced into N parts along the z -axis from the bottom of the container to height H . The height of each of these parts is

$$h_\delta = \frac{H}{N} \quad (2)$$

The volume of the n -th part is approximated using the equation below:

$$V_n \approx \frac{(S_{n-1} + S_n)h_\delta}{2} \quad (n = 1, 2, 3, \dots, N) \quad (3)$$

where S_{n-1} is the cross-sectional area of the bottom surface of the n -th part and S_n is the upper cross-sectional area. Using the definition of $S(h,\theta)$, we get:

$$S_n = S(h_\delta \cdot n, \theta) \quad (4)$$

Finally, we calculate the integration in Equation 1 by adding the volume of each part as in Equation 5:

$$V_{H,\theta} = \int_0^H S(h,\theta)dh = \sum_{n=1}^N V_n \quad (5)$$

C. Method 1: Focusing on the Target Container

Method 1 calculates the poured volume from a model of the target container.

1) *Calculation of Poured Volume based on Liquid Height:* First, we assume that the rotation angle of the target container is constant during the pouring task. Then, following the description in Section II-A, for the constant angle θ , the volume of the container under the plane of the liquid level $V_{\text{liquid},\theta}$ is the poured volume.

Finally, using the method discussed in Section II-B, $V_{\text{pour},\theta}$ is calculated using the equation below:

$$V_{\text{pour},\theta} = \int_0^H S(h, \theta) dh \quad (6)$$

The integration is calculated as described in Section II-B.

2) *PD-controller for Method 1*: The action is controlled in Method 1 using a PD-controller that takes the angular speed of the pouring container as the process variable and the poured volume as the control variable.

In Method 1, the PID-controller is used to pour the liquid precisely. It is difficult to form a complete control model of the entire pouring system if the target container is considered in isolation. However, during the pouring task, the poured liquid cannot return into the pouring container; therefore, the error in the volume V_{error} ² is always greater than 0. In other words, since the integration part of the control system is always increasing, it is of no use when controlling the pouring task. Therefore, we use the simpler PD-controller instead of a PID-controller.

The process variable, the angular speed of pouring container ω , is calculated from the proportion and differential³ of the volume error, which are expressed as below:

$$\omega = k_p V_{\text{error}} + k_d \dot{V}_{\text{error}} \quad (7)$$

A large amount of liquid can be poured in error when the V_{error} is large. We include upper and lower limits⁴ of ω to avoid this problem.

Finally, the exit condition of the pouring state is $V_{\text{error}} \leq 0$.

D. Method 2: Focus on Pouring Container

Method 2 calculates the poured volume by calculating the relation between the poured volume and the rotation angle. However, this relation is affected by the direction of the pouring container. With a cube-shaped container, for example, the relation is clearly different when pouring from the edge or from a corner. In this paper, for the sake of simplicity, we assume that the direction of pouring is constant.

Method 2 includes the following steps: First, the edge of the pouring container is detected. Second, the pouring point on this pouring edge is identified. Then, the pouring point is used to calculate the relation between the poured volume and the rotation angle. Finally, with that relation in hand, the liquid is poured to a target volume by rotating the pouring container to the target angle.

1) *Determining the Relation between Poured Volume and Rotation Angle of Pouring Container*: Method 2 depends on the following assumptions:

- 1) No spillage occurs during pouring task.
- 2) The initial volume in the pouring container is known.

²Defined by subtracting the current poured volume V_{now} from the target volume V_{tar}

³Speed of increasing volume

⁴In our program the limit is $-2^\circ/\text{s} \leq \omega \leq 2^\circ/\text{s}$

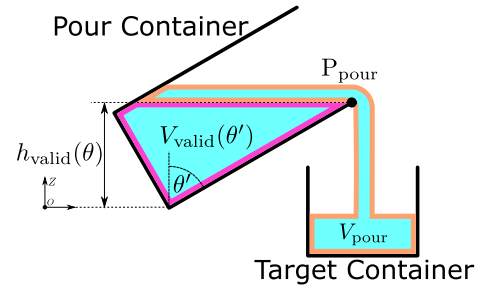


Fig. 2: This figure depicts the pouring scenario when the angle of the pouring container θ is θ' . θ' is the angle at which the liquid pours out from the pouring container, so $V_{\text{valid}}(\theta')$ (purple frame) is the volume that rests in the pouring container. $h_{\text{valid}}(\theta)$ is defined as the height between the pouring point P_{pour} and the bottom of pouring container. V_{pour} is the poured volume.

- 3) A sufficient amount of liquid is in the pouring container at the beginning of the pouring motion. ($V_{\text{init}} > V_{\text{tar}}$)
- 4) The viscosity of the liquid is low, like milk or water. The geometry of liquid deforms completely to the geometry of the container.
- 5) Surface tension is negligible.

First, the relevant variables and symbols are marked in Figure 2:

Shown in Figure 2, the relation of V_{init} , V_{pour} and $V_{\text{valid}}(\theta')$ is:

$$V_{\text{init}} = V_{\text{valid}}(\theta') + V_{\text{pour}} \quad (8)$$

The relation mentioned above implies that the relation between poured volume and the rotation angle $V_{\text{pour}}(\theta)$ is defined in the equation below (Condition (a) is $0^\circ \leq \theta \leq \theta'$, condition (b) is $\theta' < \theta \leq 180^\circ$):

$$V_{\text{pour}}(\theta) = \begin{cases} 0 & \text{if (a)} \\ V_{\text{init}} - V_{\text{valid}}(\theta) & \text{Others (b)} \end{cases} \quad (9)$$

where the calculation of $V_{\text{valid}}(\theta)$ is the same as $V_{\text{pour},\theta}$, mentioned in Section II-C.1:

$$V_{\text{valid}}(\theta) = \int_0^{h_{\text{valid}}(\theta)} S(h, \theta) dh \quad (10)$$

2) *Detection of Pouring Edge and Pouring Point*: When calculating $V_{\text{pour}}(\theta)$, the value of $h_{\text{valid}}(\theta)$ must be known. As explained in the caption of Figure 2, $h_{\text{valid}}(\theta)$ is defined as the height of the pouring point P_{pour} . This paper considers only the convex containers, so P_{pour} is defined as the lowest point of the pouring edge.

Therefore, we first need to determine the pouring edge. For any convex closed surface that is cut by a plane, the shape of the cross section can only be a closed curve, plane, straight line, or a point. However, if the surface includes a hole, the cross section can be a discontinuous curve. If the endpoints of this discontinuous curve can be identified and the surface is cut sufficiently, a point set that represents them

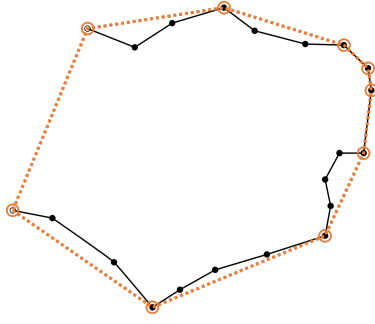


Fig. 3: Black points are the cross-sectional point set. Black circles mark the endpoints that need to be found. Points surrounded by orange circles are the results of [16], and these points are stored in an ordered array (marked with an orange broken line).

can be found, and this hole coincides with the pouring edge in this paper.

To cut the container model enough times, we set the model in several poses using the transform matrix $\mathbf{T}$. For each pose, we slice the model into several parts from bottom to top. This slicing model is described below in Section III-C. The endpoints of the discontinuous curve formed by each slice are identified. Finally, these points identified for each pose, are merged into one point set by multiplying them with the inverse matrix $\mathbf{T}^{-1}$.

To detect the pouring point, we use the detected pouring edge and transform it to several poses. The lowest point of each pose is identified, and these points are merged into a point set using the transform matrix. Finally, these points are clustered using the Euclidean clustering [15] and the pouring point is represented by the average position of cluster of maximum number of points.

To find the pouring edge using the method outlined above, the endpoints of the cross-sectional curve must be identified. We use a point cloud model, so the cross section will be made up of a set of discontinuous points. Therefore, the endpoints are one of these points and its neighbor that is separated by a distance larger than some threshold. However, the points in the point cloud model as captured by the camera are stored randomly in memory; therefore, the neighboring point along the shape of the cross section is difficult to define.

To find the neighbor in a randomly stored point set, we use the Quickhull algorithm [16] to calculate the convex hull of the cross-sectional shape. (shown in Figure 3)

With the result from this algorithm [16], we can identify a point and its convex hull neighbor, and analyze the endpoints (Figure 4).

The details of this analysis are:

- 1) Select the points within the green broken line. This green broken line is the border defined by the distance of a point less than r from the blue broken line. In Figure 4 r is 2.5 mm.
- 2) For each of these points, project it onto the blue broken line (in Figure 4 with cyan crosses).

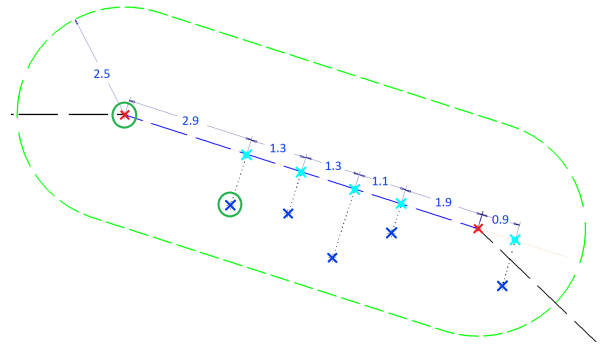


Fig. 4: This figure shows the point and its neighbor on the convex hull. Blue and black broken lines and red crosses are the results of the convex hull. Black broken lines are not related to the endpoints analysis. The green broken line is the border of points that will be analyzed. Deep blue crosses mark the inner points of the cross-sectional shape. Cyan crosses mark the inner points that will be projected onto the line defined by the two red crosses. Two points surrounded by deep green circles are the resulting endpoints.

- 3) Calculate the distance from the red and cyan crosses to their neighbor (Dimensions in Figure 4)
- 4) Select the points that lie between the red crosses. Calculate the distances from a projected point to two of these red crosses and add them up. If the sum of the distances is larger than the distance between the red crosses, that point is not between red crosses; therefore, it is ignored in the rest of the analysis.
- 5) For the rest of the points, compare the distances calculated in Step 3 with the threshold D_T . If the distance between a pair of points is larger than D_T , then that pair of points is endpoint. In Figure 4, the threshold D_T is 2.5 mm.

3) *P-Controller for Method 2*: The action is controlled in Method 2 with a P-controller that takes the angular speed of pouring container as a process variable and the rotation angle of the target volume as a control variable. The angular speed ω is determined by the difference θ_{error} between the target angle and the current angle:

$$\omega = k_p \theta_{\text{error}} \quad (11)$$

The exit condition is if the angle error is less than 0 ($\theta_{\text{error}} \leq 0$). After exiting the controller, the arm is held for t seconds so that the liquid can flow completely.

III. IMPLEMENTATION

A. Robot System and Point Cloud Model Measurement

We tested these control methods with two PA10-7C, which is a 7-degree of freedom arm made by Mitsubishi. They were set at a height of 700 mm from the ground and were separated by a distance of 804 mm. We used the Realsense D435 (RGB-D camera) made by Intel and installed one at the tip of the hand on each arm. To get a complete point cloud model of the container, we take point clouds from multiple

viewpoints in camera coordinate system. To merge these point clouds, we transform them into the world coordinate system.

B. Making the Container Model

With the merged point cloud, we can make a container model using the point cloud library (PCL) [17]), following these steps:

- 1) Noise canceling and down-sampling: For noise canceling, we use the method [15] included in PCL to remove noise far from the dense part of the point cloud. Down-sampling is performed with a voxel grid.
- 2) The table plane is removed using the random sample consensus (RANSAC) [18] method.
- 3) The rest points are clustered with the Euclidean clustering [15].
- 4) To determine the container point cloud, each cluster is filtered by the number of points and the average position of each cluster.
- 5) Then the surface of the resulting point cloud is smoothed using the moving least Squares algorithm.
- 6) Additionally, using the detection of liquid level described in Section III-D, we also calculate the axis align bounding box (AABB) of the container model in the world coordinate system.

C. Slicing the Point Cloud and Calculating Cross-sectional Area

The container model is sliced at 1 mm intervals along the z-axis. To calculate the area of each slice, the sliced points are projected onto the lower plane of the slice, and these points are regarded as the cross section. Finally, [16] is used to calculate the area of each of these cross sections.

D. Detection of Liquid Level

Method 1 needs to detect the liquid level in real time; therefore, the camera watches the target container during the pouring action. For each point cloud obtained from the camera, we remove points that are not relevant to the liquid-level detection using the AABB calculated in Section III-B. Then, with the rest of the points, we use the RANSAC to detect the plane that includes the largest number of points. Finally, the liquid level is represented by the plane detected above.

IV. EXPERIMENT

A. Environment of Experiment

1) *Containers*: In tests of these two methods, we used the three types of container shown in Figure 5. A 1000 ml milk pack was used as the pour container when testing Method 1. A dish was used as the target container when testing Method 2.

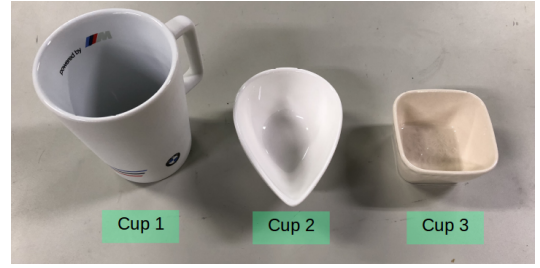


Fig. 5: Containers used in experiment. Left container is Cup 1, middle is Cup 2, right is Cup 3. In the experience at Section IV-B, each cups had poured 20 times. In the experience of Method 1, each cups had poured 10 times. In the experience of Method 2, Cup 1 was not used, Cup 2 and Cup 3 were poured 15 times each in Section IV-D, and 10 times each in Section IV-E.

2) *Ground Truth*: We use a TANITA KD-320 3 kg load scale to measure the weight of liquid in the target container. When testing volume calculation and Method 1, we used milk-tea with premeasured density (1.0276g/cm^3) as the poured liquid. When testing Method 2, we used water with density of 1.000g/cm^3 as the poured liquid.

When each pour was complete, a human experimenter read off the data displayed on the scale, which were converted to volume using the density of the liquid. Finally, the volume result was recorded.

B. Accuracy of Volume Calculation

We evaluated the accuracy of the volume calculation involved in the automated pours. Human experimenter added 50 ml milk-tea measured by scaler into Cup 1 for 4 times in each trial. In other words, in each trial, the volume was varied to 50, 100, 150, and 200 ml sequentially. In Cup 2 and 3, 25 and 15 ml milk-tea also added 4 times in each trial. For each trial, the program makes a new model of the target container, and the program calculates the volume of liquid for each adding. In the experiment, the program includes the algorithms mentioned in Sections II-B, III-A, III-C, and III-D.

Finally, we did 5 trials for each cup and the statistical results of the volume-calculation errors are shown in the Table I and Figure 6 using violin plots.

TABLE I: The average (Avg.) and standard deviation (S.d.) of errors in volume calculation

	Cup 1	Cup 2	Cup 3
Avg. (ml)	2.28	0.66	0.26
S.d. (ml)	3.29	2.94	1.44

From the results of this experiment, Cups 1 and 2 returned relatively more error than Cup 3. For Cup 1, the point cloud of the bottom of the cup is difficult to identify owing to its deep and narrow shape. The bottom of the Cup 1 model is incomplete; therefore, the error in Cup 1 is the largest. The

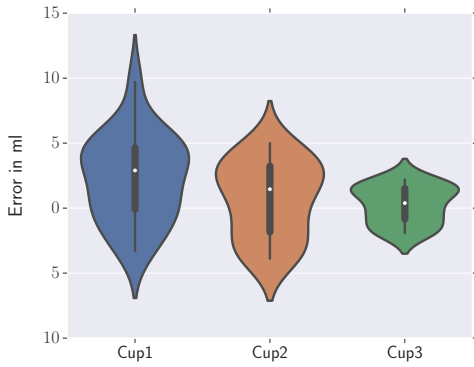


Fig. 6: Results of volume-calculation accuracy. The horizontal axis is type of cup and it also represents the error distribution of each cup. The white dot is the median of errors. The thin black line is the 95% confidence interval and the thick line is the interquartile range.

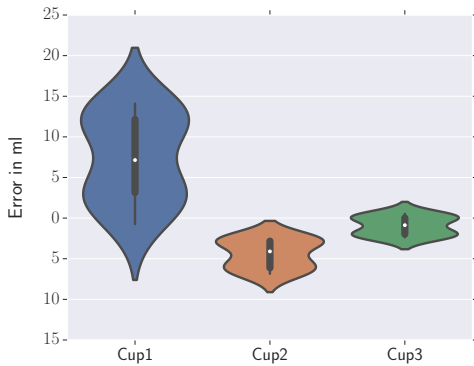


Fig. 7: The accuracy of Method 1

shape of Cup 2 is shallow so the complete model is easy to obtain but its curved surface causes distortions at the edges of the liquid-level plane. Therefore, the average height of the liquid-level plane is overestimated. The regular shape of Cup 3 explains why it gave the best performance of the three containers.

C. Method 1: Accuracy of Method 1

The accuracy of Method 1 was evaluated first. For the three types of container, we set two target volumes (Cup 1: 100 and 200 ml; Cup 2: 45 and 90 ml; and Cup 3: 30 and 60 ml). For each target volume, the arms poured liquid five times, and a new model of the target container was made for each pour.

The statistics about the Method 1 errors are shown in Table II and Figure 7 also with violin plots.

From these results, since we only selected two target volumes for this test, two peaks appear in Figure 7 and each peak represents the accuracy of the pouring action for each target volume. In this experiment, a large number of errors are contributed by Cup 1 owing to its incomplete model. The reason that the error in Cup 2 is lower than that in

TABLE II: The average (Avg.) and standard deviation (S.d.) of errors in Method 1

	Cup 1	Cup 2	Cup 3
Avg. (ml)	7.35	-4.43	-0.95
S.d. (ml)	5.45	1.79	1.16

the experiment above is because we selected only two target volumes.

D. Method 2: Influence of Surface Tension on Method 2

Before evaluating the accuracy of Method 2, we need to know how the surface tension influences the results of pouring.

First, we made a model of each pouring container. Then this model and the initial volume was input to the program for calculating the relation between poured volume and rotation angle. The Cup 1 was not used in experiment, because our robot arms are use the parallel gripper, and it can not grasp the Cup 1 stably while pouring. The initial volume in the pouring container for Cup 2 was 30, 60, and 90 ml and that for Cup 3 was 22, 44, and 66 ml. Next, the initial volume of liquid (water, in this experiment) was poured into the pouring container. Finally, the pouring container was rotated from 0° to 100° at 5° intervals. At each angle, we used the scale to measure the mass of poured liquid (water) and converted these masses to volumes. Additionally, for each initial volume, we repeat the above steps five times. Each time, the model of the pouring container is remade.

To obtain ground-truth values in the experiment above, we poured a thin layer of dishwasher detergent thinly into the pouring container to weaken the influence of surface tension.

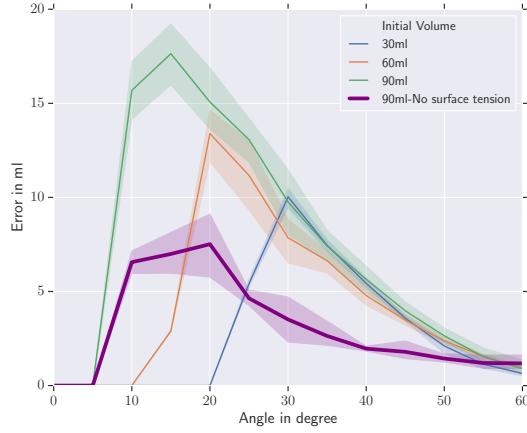
TABLE III: The average (Avg.) and standard deviation (S.d.) of errors in result of surface tension (s.t.) influence with Cup 2 in 22 ml 44 ml 66 ml and 66 ml without surface tension (66 ml no s.t.)

	Initial volume			
	22 ml	44 ml	66ml	66 ml no s.t.
Avg. (ml)	1.79	3.03	4.07	1.60
S.d. (ml)	2.32	2.32	2.97	1.27

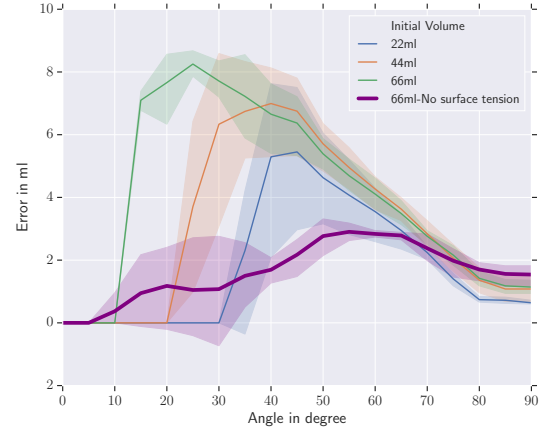
TABLE IV: The average (Avg.) and standard deviation (S.d.) of errors in result of surface tension (s.t.) influence with Cup 3 in 30 ml 60 ml 90 ml and 90 ml without surface tension (90 ml no s.t.)

	Initial volume			
	30 ml	60 ml	90ml	90 ml no s.t.
Avg. (ml)	2.75	4.24	7.18	3.03
S.d. (ml)	3.26	4.32	6.32	2.66

Results in Table III, IV and Figure 8 show that the maximum error is approximately 8 ml with Cup 2 (Cup 3



(a) Errors with Cup 2.



(b) Errors with Cup 3

Fig. 8: Influence of Surface Tension on Method 2. In Figure 8a, the blue, orange, and green lines show the results of Cup 2 with initial volumes of 30, 60, and 90 ml, respectively. Cup 3 is 22, 44, and 66 ml (Figure 8b). The purple line shows the result with no surface tension and Cup 2 with an initial volume of 90 ml.

is about 18 ml), the surface tension has a large influence and was linear after water was poured. Without surface tension, for Cup 2, the error decreased to maximum 3 ml, and was decreased to maximum 7 ml with Cup 3. The error is still large in Cup 3 without surface tension because of the large volume of water poured at the beginning of the process (approximately from 10° to 25°), and the dishwasher detergent also flowed with this water. So, the dishwasher detergent's effect on surface tension disappeared after the initial angle increment.

E. Method 2: Accuracy of Method 2

With the above-discussed results about the influence of surface tension, we can evaluate the accuracy of Method 2.

This experiment used the same steps and trails as the experiment described above. However, the initial volumes in Cup 2 and Cup 3 were fixed to 100 ml and 70 ml, respectively, and the rotation angle is calculated from the target volume using the relation between the poured volume and rotation angle. The target volumes in this experiment with Cup 2 were 0, 5, 10, ..., and 95 ml, whereas for Cup 3, they were 0, 5, 10, ..., and 65 ml. Additionally, this experiment includes the effect of surface tension.

The results of the tests with Cup 2 and Cup 3 are shown in Figure 9 and Table V.

TABLE V: The average (Avg.) and standard deviation (S.D.) of errors in Method 2

	Cup 2	Cup 3
Avg. (ml)	11.88	4.96
S.D. (ml)	4.63	2.57

From these results, with Cup 2, the maximum error is approximately 18 ml, which is the same as the results of the experiment described in Section IV-D.



Fig. 9: Errors of Method 2

F. Results and Discussions

From the results of the experiment, we found that Method 1 in Cup 1, 2, 3 had 7.35 ± 5.45 ml, -4.4 ± 1.79 ml and -0.95 ± 1.16 ml, and Method 2 in Cup 2, 3 had 11.88 ± 4.63 ml, 4.96 ± 2.57 ml mean error and standard deviation.

Compare with [12], [11], [14], [12] only pour the liquid to a specified height. Then they assume the cup is a cylinder model, and convert the error of height to the volume error, they got [13.5 ml $\pm$ 30.5 ml] error with their approach; [11] need a large amount of data to train the neural networks and hidden Markov model, the error of their approach is about 38 ml; [14] assume the target container is a cylinder shape and the pouring container is symmetric, and this approach got the pouring error is about 5 ml. In our approach, it can pour the liquid to a specified volume for any shape of containers with a high accuracy.

V. CONCLUSIONS AND FUTURE TASKS

Herein, two methods for pouring liquid precisely from a pouring container to a target container without the need of any external tools for the service robot were proposed.

- 1) Method 1 concentrates on the target container. The poured volume is calculated using a model of the target container and the liquid-level height of the poured liquid. Action is controlled using a PD-controller that takes the angular speed of the pouring container as the process variable and the poured volume as the control variable.
- 2) Method 2 concentrates on the pouring container. The poured volume is calculated using the relation between the angle of the pouring container and the poured volume. The control is simpler, only rotating the pouring container to the target angle.

Then, we tested these two methods with our dual-arm robot system. In the experiments, we implemented methods to form the model and detect the liquid level. Next, we performed four experiments using three differently shaped containers and tested the proposed method with 80 pours. From the results of the experiment, we know that Method 1 and Method 2 are enough to pour specifying volume of liquid into a unknow container with a high accuracy.

In future work, first, we will use a concave hull to process the cross sections, to extend the method so it can deal with concave-shaped containers. Second, we plan to improve on Method 2 so that it can account for the effect of surface tension.

REFERENCES

- [1] R. Mottaghi, C. Schenck, D. Fox, and A. Farhadi, "See the glass half full: Reasoning about liquid containers, their volume and content," *2017 IEEE International Conference on Computer Vision (ICCV)*, pp. 1889–1898, 2017.
- [2] C. Do, T. Schubert, and W. Burgard, "A probabilistic approach to liquid level detection in cups using an rgb-d camera," *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2075–2080, 2016.
- [3] A. Yamaguchi and C. G. Atkeson, "Stereo vision of liquid and particle flow for robot pouring," in *2016 IEEE-RAS 16th International Conference on Humanoid Robots (Humanoids)*, Nov 2016, pp. 1173–1180.
- [4] C. Schenck and D. Fox, "Towards learning to perceive and reason about liquids," in *2016 International Symposium on Experimental Robotics*, D. Kulić, Y. Nakamura, O. Khatib, and G. Venture, Eds. Cham: Springer International Publishing, 2017, pp. 488–501.
- [5] M. Tamosiunaite, B. Nemec, A. Ude, and F. Wörgötter, "Learning to pour with a robot arm combining goal and shape learning for dynamic movement primitives," *Robotics and Autonomous Systems*, vol. 59, pp. 910–922, 2011.
- [6] J. D. Langsfeld, K. N. Kaipa, R. J. Gentili, J. A. Reggia, and S. K. Gupta, "Incorporating failure-to-success transitions in imitation learning for a dynamic pouring task," 2014.
- [7] L. D. Roza, P. Jiménez, and C. Torras, "Force-based robot learning of pouring skills using parametric hidden markov models," *9th International Workshop on Robot Motion and Control*, pp. 227–232, 2013.
- [8] A. Yamaguchi, C. G. Atkeson, S. Niekum, and T. Ogasawara, "Learning pouring skills from demonstration and practice," *2014 IEEE-RAS International Conference on Humanoid Robots*, pp. 908–915, 2014.
- [9] S. Brandi, O. Kroemer, and J. Peters, "Generalizing pouring actions between objects using warped parameters," in *2014 IEEE-RAS International Conference on Humanoid Robots*, Nov 2014, pp. 616–621.
- [10] Z. Pan and D. Manocha, "Feedback motion planning for liquid pouring using supervised learning," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Sep. 2017, pp. 1252–1259.
- [11] C. Schenck and D. Fox, "Visual closed-loop control for pouring liquids," *2017 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2629–2636, 2017.
- [12] C. Do and W. Burgard, "Accurate pouring with an autonomous robot using an rgb-d camera," in *IAS*, 2018.
- [13] M. Kennedy, K. Queen, D. Thakur, K. Daniilidis, and V. Kumar, "Precise dispensing of liquids using visual feedback," *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 1260–1266, 2017.
- [14] M. Kennedy, K. Schmeckpeper, D. Thakur, C. Jiang, V. Kumar, and K. Daniilidis, "Autonomous precision pouring from unknown containers," *IEEE Robotics and Automation Letters*, vol. 4, no. 3, pp. 2317–2324, July 2019.
- [15] R. B. Rusu, "Semantic 3d object maps for everyday manipulation in human living environments," *KI - Künstliche Intelligenz*, vol. 24, pp. 345–348, 2009.
- [16] C. B. Barber, D. P. Dobkin, and H. Huhdanpaa, "The quickhull algorithm for convex hulls," *ACM Trans. Math. Softw.*, vol. 22, pp. 469–483, 1996.
- [17] R. B. Rusu and S. Cousins, "3d is here: Point cloud library (pcl)," *2011 IEEE International Conference on Robotics and Automation*, pp. 1–4, 2011.
- [18] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Commun. ACM*, vol. 24, pp. 381–395, 1981.